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| <b>CONTACT INFORMATION</b> | Po-An Tsai<br>2 Technology Park Dr, Westford, MA 01886                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | poant@nvidia.com<br><a href="https://research.nvidia.com/person/po-an-tsai">https://research.nvidia.com/person/po-an-tsai</a> |
| <b>RESEARCH INTERESTS</b>  | <b>Computer systems and architecture; machine learning and tensor algebra accelerator architectures.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                               |
| <b>EDUCATION</b>           | <b>Ph.D. in Computer Science</b> , June 2019 <i>Massachusetts Institute of Technology</i> <ul style="list-style-type: none"> <li>• Advisor: Professor Daniel Sanchez</li> <li>• Thesis: Redesigning the Memory Hierarchy to Exploit Static and Dynamic Application Information</li> </ul> <b>S.M. in Computer Science</b> , June 2015 <i>Massachusetts Institute of Technology</i><br><b>B.S. in Electrical Engineering</b> , June 2012 <i>National Taiwan University (NTU), Taiwan</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                               |
| <b>HONORS AND AWARDS</b>   | <b>Best Paper Award</b> , ISCA-53, 2026<br><b>IEEE Micro Top Picks Honorable Mention</b> , 2025<br><b>Best Paper Award Nominee</b> , ISCA-51, 2024<br><b>Distinguished Artifact Award</b> , MICRO-55, 2022<br><b>IEEE Micro Top Picks Award</b> , 2021<br><b>Best Paper Award Nominee</b> , HPCA-21, 2015<br><b>Best Poster Award</b> , MIT Industry-Academia Partnership Workshop, 2014<br><b>Jacobs Presidential Fellowship</b> , MIT, 2013<br><b>Valedictorian</b> , NTUEE, 2012<br><b>Presidential Award</b> , NTU, 2010, 2011, 2012<br><b>Second Prize</b> , NTUEE Undergraduate Research Award, 2012<br><b>Star Futures Award</b> , Altera International FPGA Design Contest, 2011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                               |
| <b>WORK EXPERIENCE</b>     | <b>Principal Research Scientist</b> , March 2026 – Current <i>Architecture Research Group, NVIDIA</i><br><b>Sr. Research Scientist</b> , July 2022 – February 2026<br><b>Research Scientist</b> , July 2019 – June 2022<br><b>Research Assistant</b> , September 2013 – May 2019 <i>Computation Structures Group, MIT</i><br><b>Software Engineering Intern</b> , Summer 2015 <i>Distributed Resource Management Team, VMware</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                               |
| <b>PUBLICATIONS</b>        | <b>AMMA: A Multi-Chiplet Memory-Centric Architecture for Low-Latency 1M Context Attention Serving</b><br>Zhongkai Yu, Haotian Ye, Chenyang Zhou, Ohm Rishabh Venkatachalam, Zaifeng Pan, Zhengding Hu, Junsung Kim, Won Woo Ro, <b>Po-An Tsai</b> , Shuyi Pei, Yangwook Kang, Yufei Ding<br>The 59th Annual International Symposium on Microarchitecture (MICRO-59), November 2026.<br><b>Patterns behind Chaos: Forecasting Data Movement for Efficient Large-Scale MoE LLM Inference</b><br>Zhongkai Yu, Yue Guan, Zihao Yu, Chenyang Zhou, Zhengding Hu, Shuyi Pei, Yangwook Kang, Yufei Ding, <b>Po-An Tsai</b><br>The 53rd Annual International Symposium on Computer Architecture (ISCA-53), June 2026.<br><b>RocketKV: Accelerating Long-context LLM Inference via Two-stage KV Cache Compression</b><br>Payman Behnam, Yaosheng Fu, Ritchie Zhao, <b>Po-An Tsai</b> , Zhiding Yu, Alexey Tumanov<br>Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), July 2025.<br><b>Abstracting Sparse DNN Acceleration via Structured Sparse Tensor Decomposition</b><br>Geonhwa Jeong, <b>Po-An Tsai</b> , Abhimanyu R. Bambhaniya, Stephen W. Keckler, Tushar Krishna<br>The 8th Annual Conference on Machine Learning and Systems (MLSys 2025), May 2025.<br><b>Sparsepipe: Sparse Inter-operator Dataflow Architecture with Cross-Iteration Reuse</b><br>Yunan Zhang, <b>Po-An Tsai</b> , Hung-Wei Tseng<br>The 57th Annual International Symposium on Microarchitecture (MICRO-57), November 2024.<br><b>Mind the Gap: Attainable Data Movement and Operational Intensity Bounds for Tensor Algorithms</b><br>Qijing (Jenny) Huang, <b>Po-An Tsai</b> , Joel S Emer, Angshuman Parashar |                                                                                                                               |

- The 51st Annual International Symposium on Computer Architecture (ISCA-51), June 2024.  
**Best Paper Award Nominee.**
- SDQ: Sparse Decomposed Quantization for LLM Inference**  
 Geonhwa Jeong, **Po-An Tsai**, Stephen W. Keckler, Tushar Krishna  
 arXiv preprint arXiv:2406.13868, June 2024.
- Symphony: Orchestrating Sparse and Dense Tensors with Hierarchical Heterogeneous Processing**  
 Michael Pellauer, Jason Clemons, Vignesh Balaji, Neal Crago, Aamer Jaleel, Donghyuk Lee, Mike O'Connor, Angshuman Parashar, Sean Treichler, **Po-An Tsai**, Stephen W. Keckler, Joel S. Emer  
 ACM Transactions on Computer Systems, Vol. 41, December 2023.
- RM-STC: Row-Merge Dataflow Inspired GPU Sparse Tensor Core for Energy-Efficient Sparse Acceleration**  
 Guyue Huang, Zhengyang Wang, **Po-An Tsai**, Chen Zhang, Yufei Ding, Yuan Xie  
 The 56th IEEE/ACM International Symposium on Microarchitecture (MICRO-56), October 2023.
- HighLight: Efficient and Flexible DNN Acceleration with Hierarchical Structured Sparsity**  
 Yannan Nellie Wu, **Po-An Tsai**, Saurav Muralidharan, Angshuman Parashar, Vivienne Sze, Joel S. Emer  
 The 56th IEEE/ACM International Symposium on Microarchitecture (MICRO-56), October 2023.
- Accelerating Sparse Data Orchestration via Dynamic Reflexive Tiling**  
 Toluwanimi O. Odemuyiwa, Hadi Asghari-Moghaddam, Michael Pellauer, Kartik Hegde, **Po-An Tsai**, Neal C. Crago, Aamer Jaleel, John D. Owens, Edgar Solomonik, Joel S. Emer, Christopher W. Fletcher  
 The 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS-28), March 2023.
- Demystifying Map Space Exploration for NPUs**  
 Sheng-Chun Kao, Angshuman Parashar, **Po-An Tsai**, Tushar Krishna  
 2022 IEEE International Symposium on Workload Characterization (IISWC), November 2022.
- Sparseloop: An Analytical Approach To Sparse Tensor Accelerator Modeling**  
 Yannan Nellie Wu, **Po-An Tsai**, Angshuman Parashar, Vivienne Sze, Joel S. Emer  
 The 55th IEEE/ACM International Symposium on Microarchitecture (MICRO-55), October 2022.
- Distinguished Artifact Award.**
- SIMD<sup>2</sup>: A Generalized Matrix Instruction Set for Accelerating Tensor Computation beyond GEMM**  
 Yunan Zhang, **Po-An Tsai**, Hung-Wei Tseng  
 The 49th Annual International Symposium on Computer Architecture (ISCA-49), June 2022.
- Ruby: Improving Hardware Efficiency for Tensor Algebra Accelerators Through Imperfect Factorization**  
 Mark Horeni, Pooria Taheri, **Po-An Tsai**, Angshuman Parashar, Joel S. Emer, Siddharth Joshi  
 IEEE International Symposium on Performance Analysis of Systems and Software, May 2022.
- Union: A Unified HW-SW Co-Design Ecosystem in MLIR for Evaluating Tensor Operations on Spatial Accelerators**  
 Geonhwa Jeong, Gokcen Kestor, Prasanth Chatarasi, Angshuman Parashar, **Po-An Tsai**, Sivasankaran Rajamanickam, Roberto Gioiosa, and Tushar Krishna  
 The 30th International Conference on Parallel Architectures and Compilation Techniques (PACT-30), September 2021.
- Leaking Secrets through Compressed Caches**  
**Po-An Tsai**, Andres Sanchez, Christopher W. Fletcher, and Daniel Sanchez.  
 IEEE Micro's Top Picks from the Computer Architecture Conferences, May/June 2021.
- Mind Mappings: Enabling Efficient Algorithm-Accelerator Mapping Space Search**  
 Kartik Hegde, **Po-An Tsai**, Sitao Huang, Vikas Chandram, Angshuman Parashar, and Christopher W. Fletcher.  
 The 25th International Conference on Architectural Support for Programming Languages and Operating Systems, (ASPLOS-26), April 2021.
- Sparseloop: An Analytical, Energy-Focused Design Space Exploration Methodology for Sparse Tensor Accelerators**  
 Yannan Nellie Wu, **Po-An Tsai**, Angshuman Parashar, Vivienne Sze, Joel S. Emer  
 IEEE International Symposium on Performance Analysis of Systems and Software, March 2021.
- Hardware Abstractions for Targeting EDDO Architectures with the Polyhedral Model**  
 Angshuman Parashar, Prasanth Chatarasi, and **Po-An Tsai**.  
 International Workshop on Polyhedral Compilation Techniques (IMPACT), January 2021.

**Safecracker: Leaking Secrets through Compressed Caches**

**Po-An Tsai**, Andres Sanchez, Christopher W. Fletcher, and Daniel Sanchez.

The 25th International Conference on Architectural Support for Programming Languages and Operating Systems, (ASPLOS-25), March 2020.

**Compress Objects, Not Cache Lines: An Object-Based Compressed Memory Hierarchy**

**Po-An Tsai** and Daniel Sanchez.

The 24th International Conference on Architectural Support for Programming Languages and Operating Systems, (ASPLOS-24), April 2019.

**Rethinking the Memory Hierarchy for Modern Languages**

**Po-An Tsai**, Yee Ling Gan, and Daniel Sanchez.

The 51st International Symposium on Microarchitecture (MICRO-51), October 2018.

**Adaptive Scheduling for Systems with Asymmetric Memory Hierarchies**

**Po-An Tsai**, Changping Chen, and Daniel Sanchez.

The 51st International Symposium on Microarchitecture (MICRO-51), October 2018.

**KPart: A Hybrid Cache Partitioning-Sharing Technique for Commodity Multicores**

Nosayba El-Sayed, Anurag Mukkara, **Po-An Tsai**, Harshad Kasture, Xiaosong Ma, and Daniel Sanchez.

The 24th Intl. Symposium on High Performance Computer Architecture (HPCA-24), February 2018.

**Nexus: A New Approach to Replication in Distributed Shared Caches**

**Po-An Tsai**, Nathan Beckmann, and Daniel Sanchez.

The 26th International Conference on Parallel Architectures and Compilation Techniques (PACT-26), September 2017.

**Jenga: Software-Defined Cache Hierarchies**

**Po-An Tsai**, Nathan Beckmann, and Daniel Sanchez.

The 44th International Symposium on Computer Architecture (ISCA-44), June 2017.

**Scaling Distributed Cache Hierarchies with Computation and Data Co-Scheduling**

Nathan Beckmann, **Po-An Tsai**, and Daniel Sanchez.

The 21st International Symposium on High Performance Computer Architecture (HPCA-21), February 2015.

**Best Paper Award Nominee.**

**Hybrid Path-Diversity-Aware Adaptive Routing with Latency Prediction Model in Network-on-Chip Systems**

**Po-An Tsai**, Yu-Hsin Kuo, En-Jui Chang, and An-Yeu Wu.

International Symposium on VLSI Design, Automation & Test, (VLSI-DAT), March 2013.

**Path-Diversity-Aware Adaptive Routing in Network-on-Chip Systems**

Yu-Hsin Kuo, **Po-An Tsai**, Hao-Ping Ho, En-Jui Chang, Hsien-Kai Hsin, and An-Yeu Wu.

The 6th International Symposium on Embedded Multicore SoCs (MCSoc), September 2012.

**PATENT****Resource-Based Virtual Computing Instance Scheduling**

US Patent 15283274

**Flexible Accelerator for a Tensor Workload**

US Patent 17343597, 17343582

**Pruning and accelerating neural networks with hierarchical fine-grained structured sparsity**

US Patent 17681967

**Generating sparse neural networks**

US Patent 18222916

**Method and apparatus for direct convolution calculation**

US Patent 18449381

**Compression of machine learning models via sparsification and quantization**

US Patent 18602951

**Expressive sparse matrix representations with limited metadata**

US Patent 18777367, 18777381, 18777519, 18777379

**SERVICE**

- Area Chair: MICRO'26
- Program Committee Member: ISMM'20, HPCA'21, MICRO'22, DAC'24, MICRO'24, ISCA'25, MICRO'25, ISCA'26, IEEE MICRO Top Picks'26.

- External Review Committee Member: PACT'20, MICRO'20, MICRO'21, ASPLOS'22, ISCA'22, ISCA'23, ISCA'24, ASPLOS'27.
- Reviewer: ACM TACO, ACM TCAD, IEEE JSSC, IEEE CAL.
- ACM Student Research Competition Co-Chair: MICRO'24.
- Artifact Evaluation Co-Chair: ISCA'23.
- Workshop/Tutorial Co-Chair: HPCA'21.
- Submissions Co-Chair: MICRO'17.
- Organizer: Workshop on Democratizing Domain-Specific Accelerators @ MICRO'22, MICRO'23, MICRO'24.
- Organizer: Tutorial on Sparse Tensor Accelerators: Abstraction and Modeling @ ISCA'21.
- Organizer: Tutorial on Timeloop/Accelergy: Tools for Evaluating Deep Neural Network Accelerator Designs @ MICRO'19, ISCA'20, ISPASS'20.